



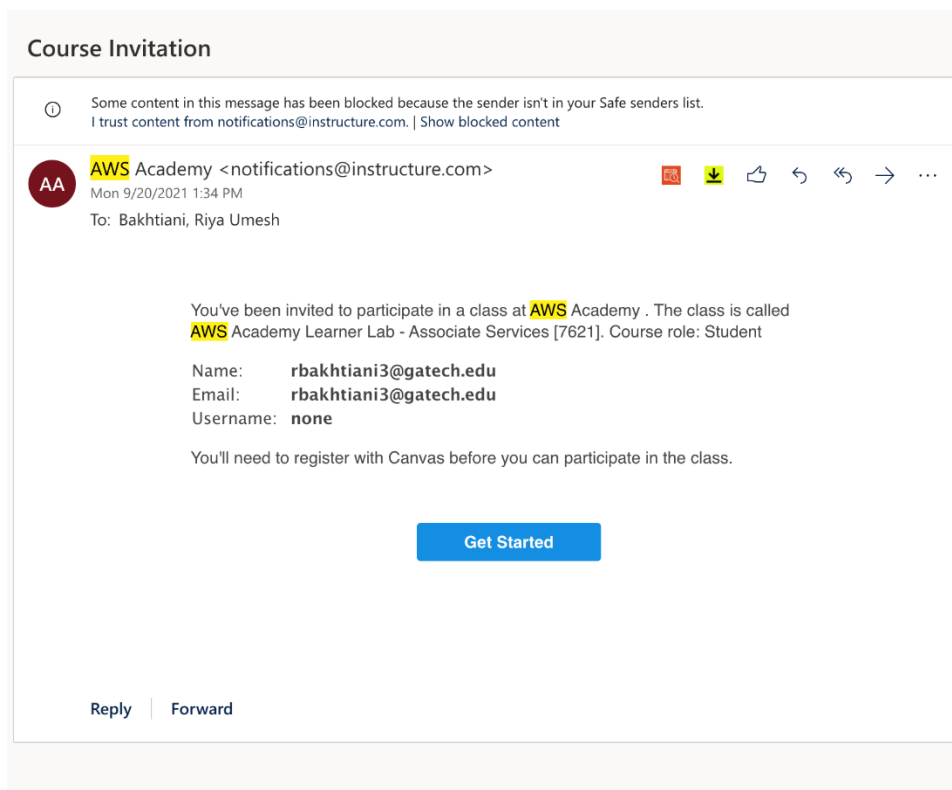
Setup Guide [For Q3]

Getting Started

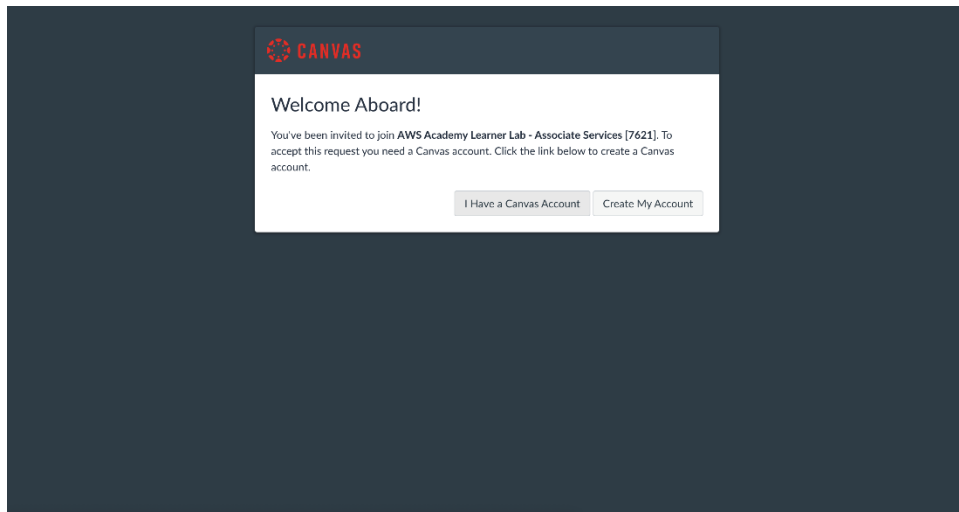
1. Create an AWS Academy account

You will receive an email from notifications@instructure.com inviting you to participate in the **AWS Academy Learner Lab – Associate Services** course. Your AWS Academy allows you to access EC2, Elastic MapReduce and S3 storage. Click on the button to **Join AWS Academy** in the email to proceed.

Note: AWS Academy does NOT work on Safari. Use a Chrome-based browser for the best results.

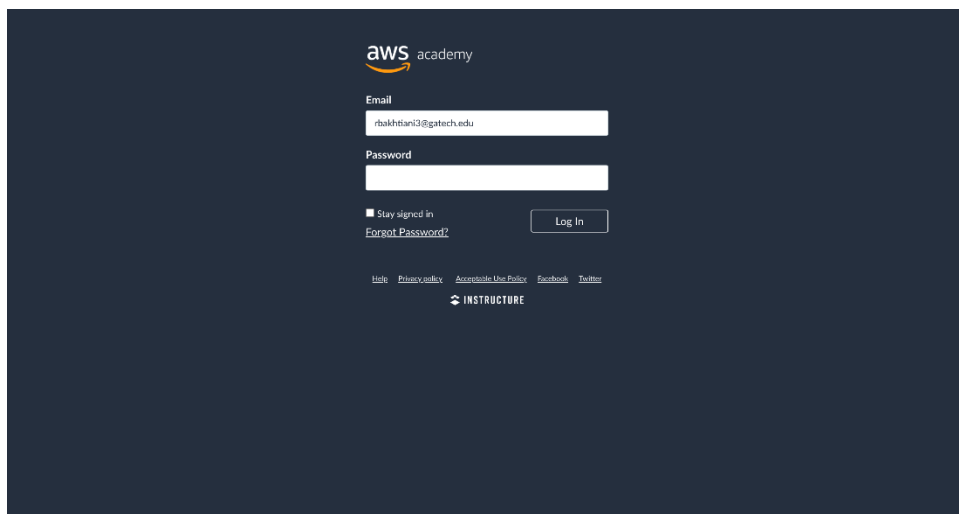


You will be taken to the Canvas page. When it appears, click on the “Create My Account” button.

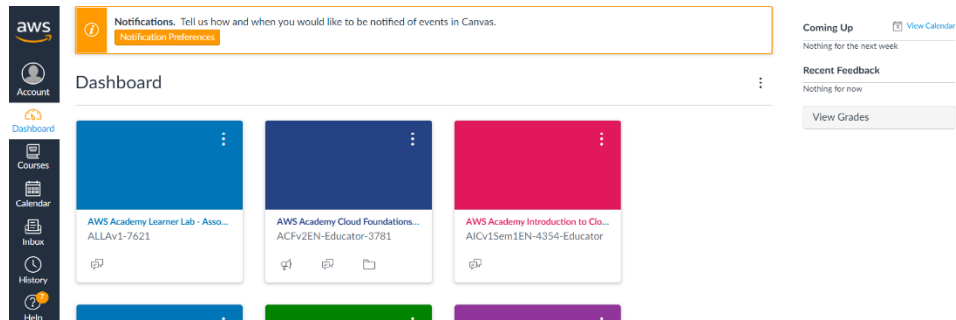


Now, fill in the requested information (e.g. e-mail address, password, etc.). Once submitting, you'll be able to log in to your account at the following URL: <https://awsacademy.instructure.com/login/canvas>

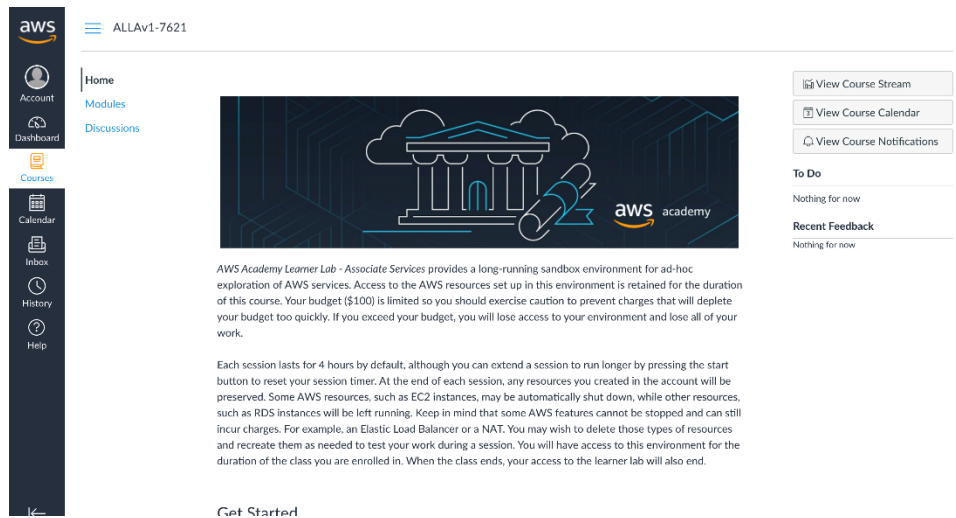
When you arrive at the login URL you will see a screen like this:



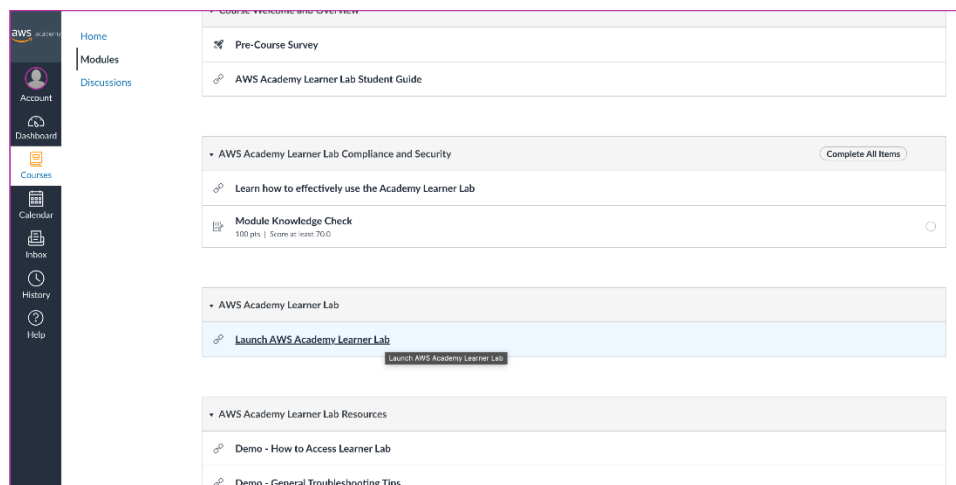
Please enter your e-mail address and password used when setting up the AWS Canvas account (not the Canvas account for the CSE 6242 course). Click the “Log In” button. You will then see the AWS Canvas home screen. Click on the “Dashboard” tab on the far left. You should see a screen like this one (although the number of courses on your screen will differ):



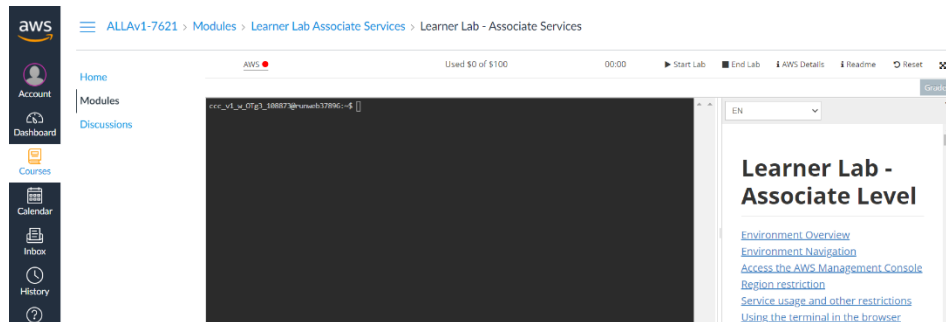
Click on the “AWS Academy Learner Lab” course button. You should see the following screen (the course homepage).



Click on the “Modules” link on the near-left menu. A list of course modules will appear:



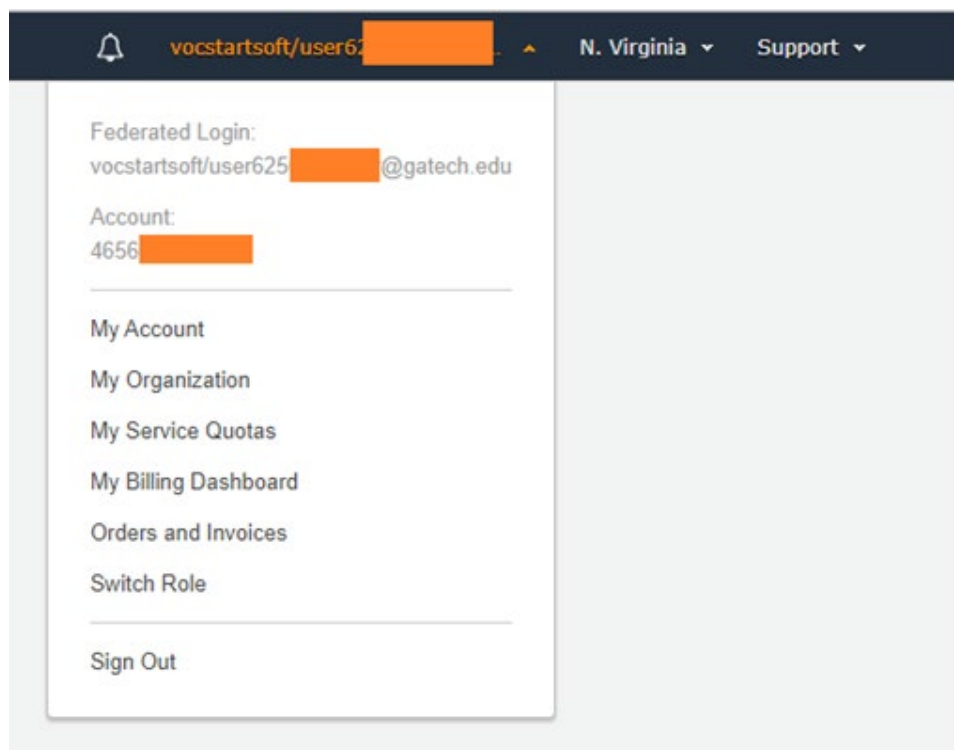
Click on the “Launch AWS Academy Learning Lab”. You’ll see a screen like the following screen:



Next, click on the “Start Lab” button on the top right. You’ll see an agreement from Vocareum. Scroll to the bottom of the agreement and click “I agree.” Next, click “Start Lab” at the top. It will take a few minutes for the account to be set up. Once the red circle next to AWS turns green in top left, click the Green button and a new window will open. Do take time to



You are now in the AWS management console! It will look something like this [right top corner].



If you have any problems with this process, please let the course staff know on EdStem via the dedicated AWS Setup thread.

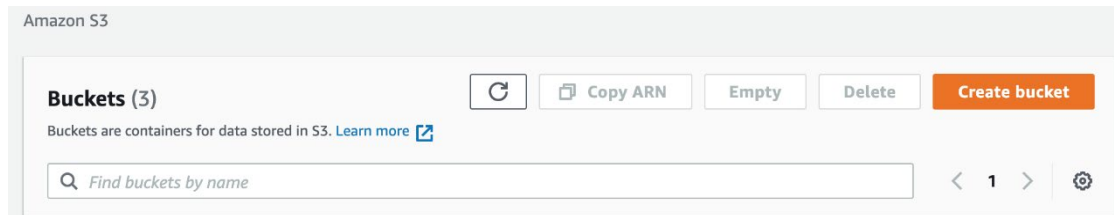
2. Create storage buckets on S3

We need S3 for storing the final results of the assignment which will be submitted to the autograder.

Data (or objects) in S3 are stored in what we call “**buckets**”. You can think of buckets as folders. All S3 buckets need to have unique names.

1. In the AWS Management Console click on **S3** under **All services** → **Storage**.

In the S3 console, click on **Create Bucket**.



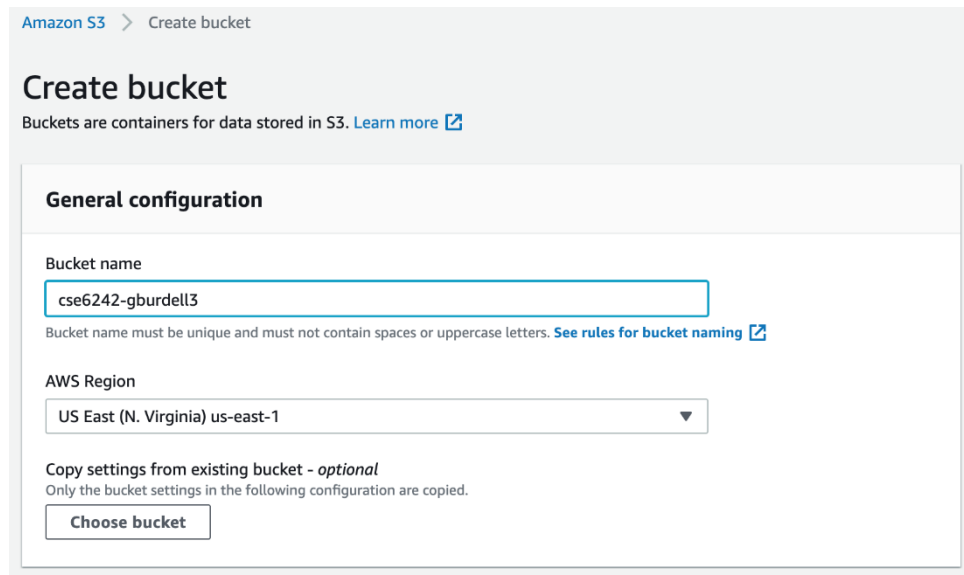
2. Create a bucket: Enter the following details (bucket name and region) then click **Create Bucket** at the bottom of the screen. **Keep all other settings the same.**

Bucket Name Format: cse6242-<GT username>

Example: cse6242-gburdell3

Region: US East (N. Virginia)

VERY IMPORTANT: Please select “**US East (N. Virginia)**” only. If you have buckets in other regions, data transfer charges would apply.



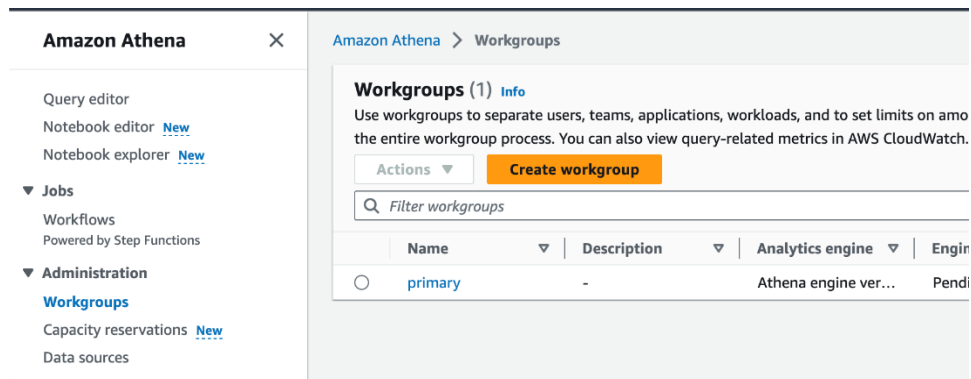
3. A new bucket will appear in the S3 console. Clicking on it will show you that it is empty.

We are done with creating S3 buckets at this point.

3. Create an Athena Workgroup

This section will cover creating a Workgroup in Amazon Athena. For further information about workgroups in Athena, click [here](#).

1. Go to Amazon Athena. Select Workgroups on the left menu. Click “Create workgroup”.



2. Make sure the region specified in the top-right corner of the page is **N. Virginia**. Otherwise click on it and from the drop-down choose N. Virginia.
3. We will now fill out the various configuration fields to create a new Workgroup:
 - a. Give your workgroup a name. It can be anything you want.

Workgroup details
Enter a unique name for your workgroup. To change the workgroup name, delete the workgroup and recreate it with a new name.

Workgroup name

hw3-q3

Workgroup name must be from 1-128 characters and must be unique per region of your account. Valid characters are a-z, A-Z, 0-9, _(underscore), .(period) and -(hyphen). This value cannot be changed after creation.

Description - optional

Workgroup description must be from 1-1024 characters. 1024 characters remaining.

- b. Select “Apache Spark” as the Analytics Engine.

Analytics engine - new Info

Choose the type of engine

☐ Athena SQL
Athena SQL engine for interactive queries.

☒ Apache Spark
Highly-scalable serverless Spark engine.

Engine version

PySpark engine version 3

☐ Turn on example notebook
Athena will upload an example notebook to your workgroup and attach additional permissions to your service role.
[View additional permissions details](#)

- c. Under Additional Configurations, choose “LabRole” as the existing IAM service role.

▼ IAM role configuration

Notebooks require permissions to call other AWS services on your behalf.

Service role to authorize Athena

☐ Create and use a new service role
☒ Use an existing service role

Service role

LabRole

[View this Service role in IAM console](#)

Notebook and calculation code encryption key management

Info

☒ Encrypt using AWS owned key (Default)

A key that AWS owns and manages for you.

☐ Encrypt using customer managed key

Choose a key you have permission to use, or create a new one.

d. Select “Create workgroup” to finish.

4. Import the hw3-q3 skeleton notebook.

4. Get started with the skeleton

In this section we will import the skeleton file to the workgroup and run our first cell.

1. Select your newly created workgroup from the list and then click “Import file”. Import the q3.ipynb file provided in the skeleton.

Amazon Athena > Workgroups > hw3-q3

hw3-q3

Delete
Turn off workgroup
Edit

To grant access to the workgroup, [create an IAM policy](#) and attach it to a user, group, or role. [Learn more](#)

Overview details

Workgroup name hw3-q3	Description -	Content encryption AWS owned
Analytics engine PySpark engine version 3	Calculation results location s3://202667119616-us-east-1-athena-results-bucket-o0pe194bbj	Result encryption -
Override client side settings N/A	Role ARN arn:aws:iam::202667119616:role/LabRole	Publish metrics to AWS CloudWatch Turned on
Workgroup status Turned on	Workgroup ARN arn:aws:athena:us-east-1:202667119616:workgroup/hw3-q3	Created on 2023-06-28T20:58:34.968-04:00

Notebooks Sessions Tags

Notebooks (0) Info

Q Search notebooks

Name

Last edited

No notebooks

Actions

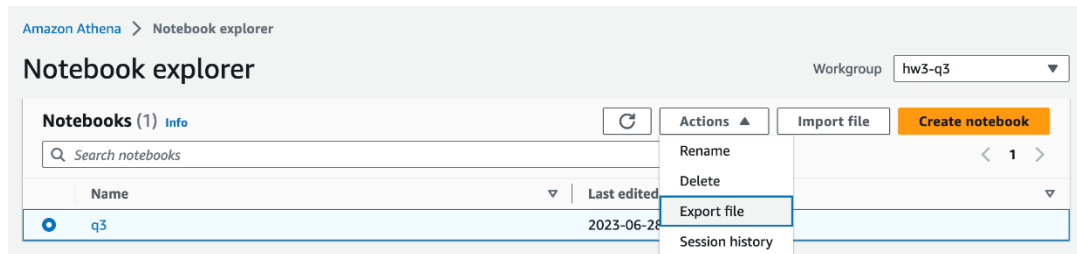
Import file

Create notebook

7

Version 0

2. Run the first code cell which will import PySpark, followed by a cell which should contain `sc` to start the Spark Application so you can start programming the assignment.
3. Once you have finished coding, you can export the notebook from “Notebook Explorer” by selecting your notebook and clicking “Export File” from the Actions dropdown.



Troubleshooting

1. If you get an error saying your credentials are invalid, when logging in



Click the logout link and then try again. If that still does not work, try clearing your cache and cookies in your browser. You can also use Incognito mode in your browser or a completely different browser if none of those options work.

2. Take time to read the Learner Lab documentation on the Lab Screen! It is full of helpful information.

